Jamovi Handbook

Contents

Welcome

Welcome to the STAT 301/307 jamovi handbook guide. This guide is designed to help you use jamovi for your homework through examples using real data.

I

Chapter 1: Getting Started

1. Installing jamovi

1.1 What is jamovi?

jamovi is a statistical software that we will use for homework assignments in this class. You can think of it like a fancy calculator that can read in data and quickly calculate statistics, summarize data, and create visualizations.

1.2 Where do I get jamovi?

Start by going to jamovi.org. After scrolling down a bit, you should see something like the figure below:

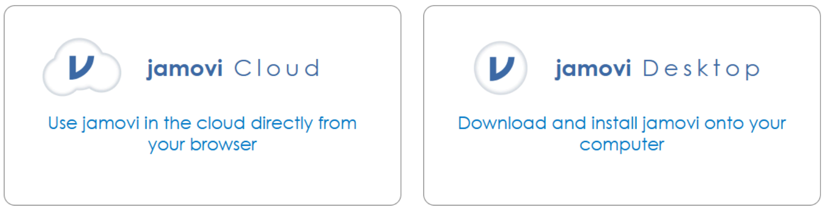


Figure 1.1

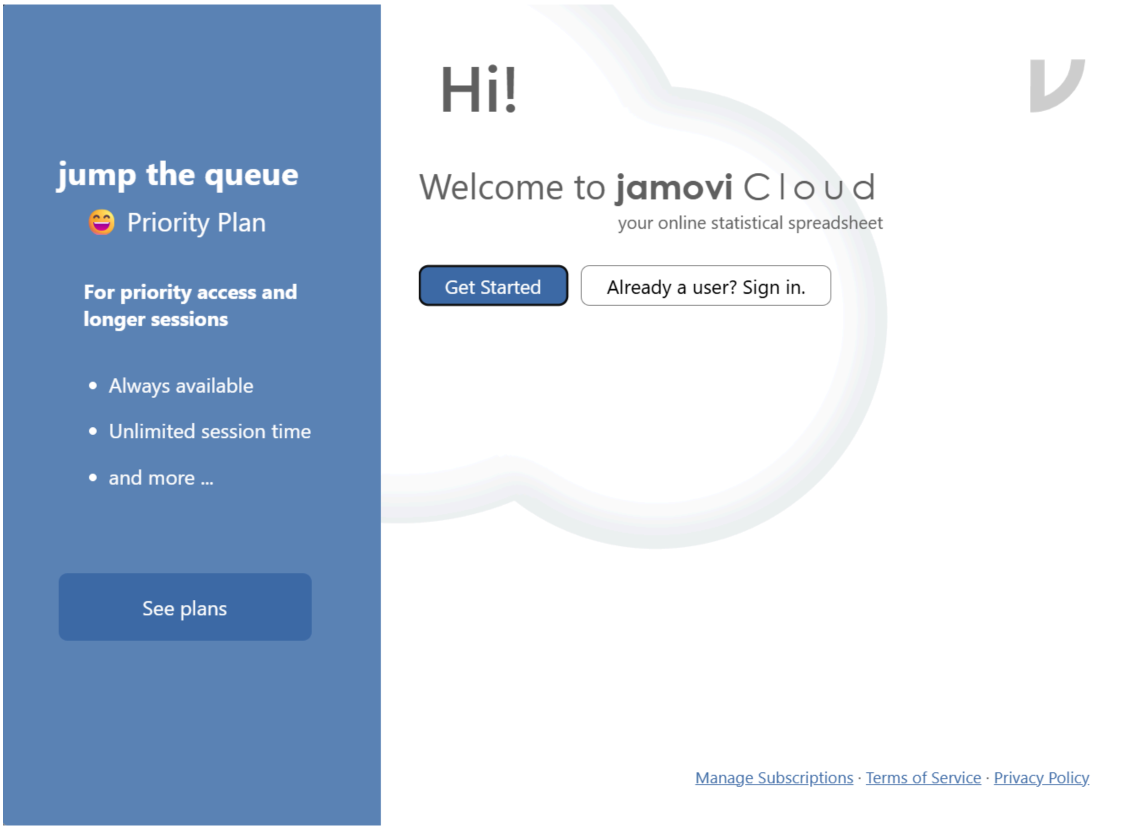
The option on the left is **jamovi Cloud**, which allows you to use jamovi in your browser without downloading or installing anything. The option on the right is **jamovi Desktop**, which downloads and installs jamovi onto your computer.

1.3 Should I use jamovi Cloud or jamovi Desktop?

In this class, you can use either. The benefit of jamovi Cloud is that you don't need to download or install anything. The downsides are that you can only use jamovi Cloud when you have an internet connection, you have a limited number of minutes per session, and you'll need to sign up for an account. We would recommend jamovi Desktop, but feel free to use jamovi Cloud if you're having trouble with the download and installation steps.

jamovi Cloud

If you opt to use jamovi Cloud, all you need to do is click on 'jamovi Cloud', then click on 'Start', which should open up a screen like this:



You'll need to start by creating an account.

1. Click on 'Get Started' –> 'Sign in as Guest' –> 'Sign in with email'.
2. Enter your CSU email, click on 'Next', and then set your password.
3. Click on 'Send verification' and click the link that is emailed to you to verify your account. It may take a few minutes for it to show up in your inbox.

> ⚠ Check your spam folder!
>
> Sometimes the verification emails can get filtered; make sure to check your spam/junk folder for the verification email.

jamovi Desktop

Download for Windows

After clicking on jamovi Desktop from the page seen in Figure 1.1, you should see the following screen, or something similar:

Figure 1.2

Click on whatever the "solid & current" version is (the number may be different than what you see in Figure 1.2) to download jamovi.

Download for Mac

After clicking on jamovi Desktop from the page seen in Figure 1.1, scroll down until you see something like this:

All Releases

OS	Release	Version	Arch	Format
Windows	solid	2.7.33	x64	.exe
				.msix
				.zip
	legacy	2.6.44	x64	.exe
				.zip
macOS	solid	2.7.32	arm64	.dmg
				.pkg
			x64	.dmg
				.pkg
	legacy	2.6.45	arm64	.dmg
			x64	.dmg
Linux		2.7.32	x64	(flathub)
			arm64	(flathub)
ChromeOS		2.7.32		(flathub)

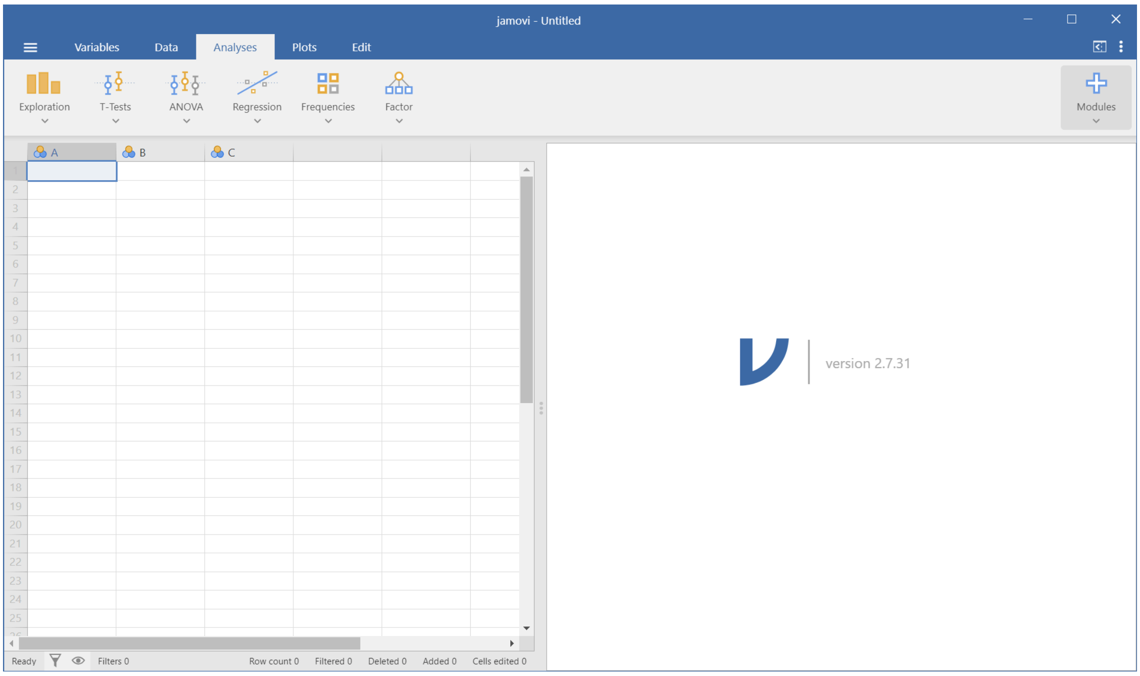
Figure 1.3

Click on whatever the 'solid' version for Mac is to download it.

Regardless of whether you have a Mac or Windows computer, double click on the file after it finishes downloading to open the installation wizard and follow the default steps to install jamovi. After it finishes installing, locate the app and start it.

2. Setting up jamovi and Reading Data

After installing jamovi and opening it up, you should see a screen without any data in it, like the image below:



The **VERY FIRST** thing we need to do is to change the number formatting:

1. Click on the three vertical dots in the very top right corner of the screen
2. Click on the dropdown menus for 'Number format' and 'p-value format', and set them *both* to **3 dp**. Make sure they are *not* set to **3 sf** or anything else.

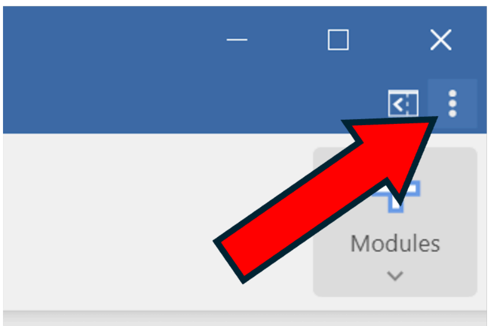


Figure 2.1: Step 1: Click on the three vertical dots.

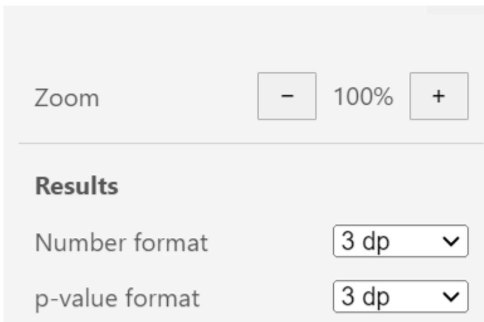
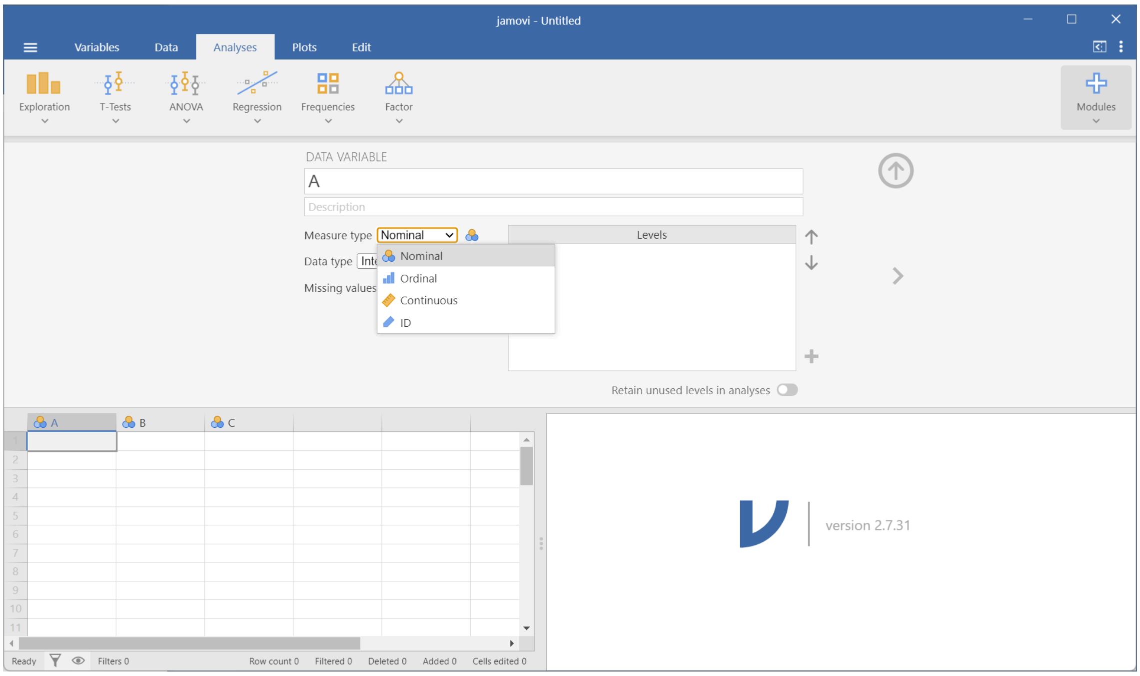


Figure 2.2: Step 2: Set the formatting to 3 dp.

To do analysis in jamovi, we'll first need data! We can either read data in via a file, or we can enter it in manually.

2.1 Entering Data Manually

To enter in data manually, you can directly type in numbers or text into each cell like in an Excel sheet. To change the name of the variable and the data type, double click the top of the column (where it has the label 'A'). It should bring up the following menu:



At the top, you can change the name of the variable to something other than 'A', and add an optional description. You can also change the variable type; here 'Nominal' is equivalent to what we refer to as **Categorical**, and 'Continuous' is equivalent to what we call **Quantitative**. They have a third option called 'Ordinal', which we don't have to worry about, and a fourth option called 'ID', which is typically for unique identifiers like names or ID numbers.

2.2 Reading in Data

In this course, data is typically in a "Comma-Separated Values" file or **CSV** file for short, which stores data in a table format. Basically all of the data we work with in this course will be stored in CSV files.

Start by downloading the **wingspan_data.csv** file from Canvas. Then in jamovi:

1. Click on the three horizontal lines in the top left corner.
2. Click 'Open'
    a. (If you're using jamovi Desktop) Click 'Browse'
    b. (If you're using jamovi Cloud) Click 'This Device'
3. This will bring up a file explorer window. Locate the **wingspan_data.csv** file you just downloaded and click 'Open'.

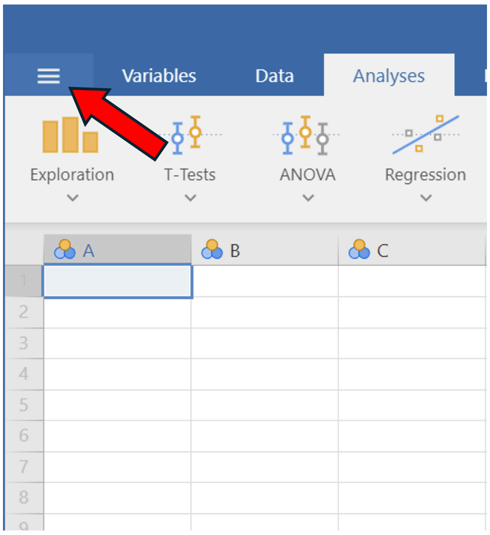


Figure 2.3: Step 1: Click on the three horizontal lines.

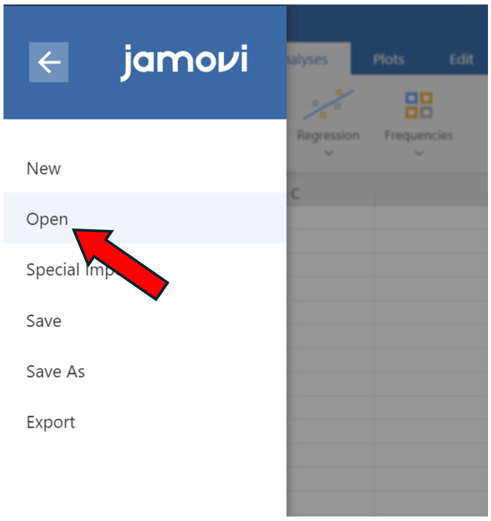
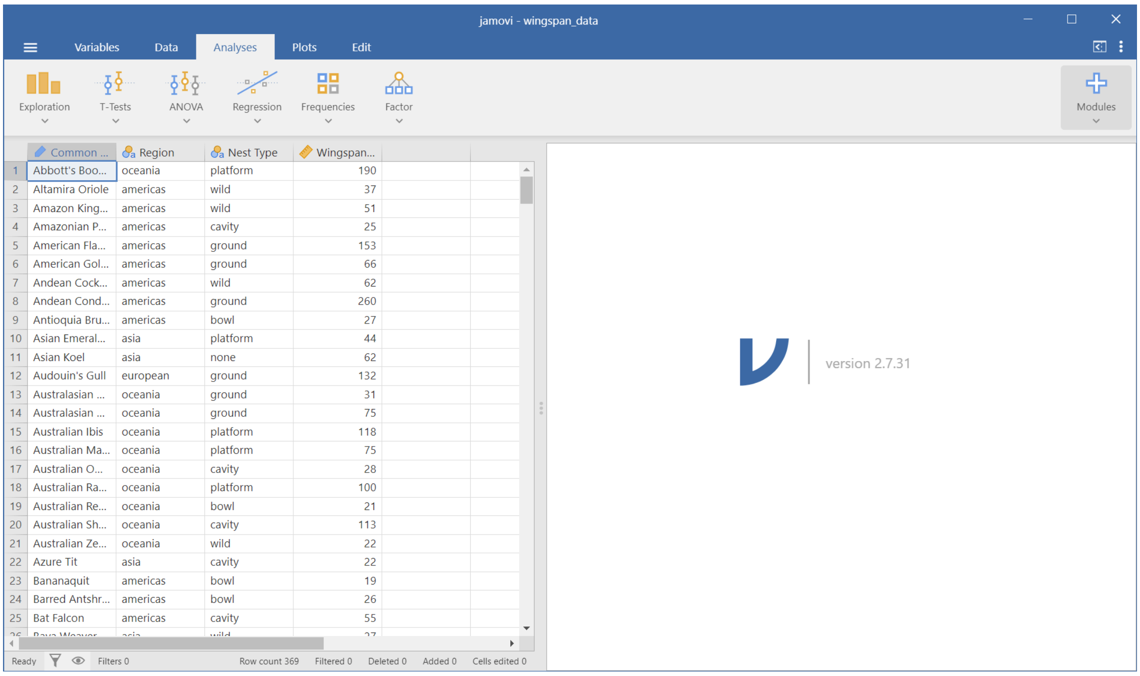


Figure 2.4: Step 2: Click on 'Open'.

Your jamovi should then look like the following:



Great! Now we have data that we will do analysis on in the next chapter.

II

Chapter 2: Describing Data

3. Descriptive Statistics

In this section we'll go over how to calculate means, medians, quartiles, and other descriptive statistics for data. Begin by loading in the **wingspan_data.csv** dataset into jamovi.

3.1 Default statistics

We'll begin by getting descriptive statistics for the wingspans of the birds.

1. Click on the 'Analyses' tab at the top of the window.
2. Click on 'Exploration' –> 'Descriptives'
3. To get statistics for the wingspans, click on the **Wingspan** variable on the left, then click on the arrow adding it to the 'Variables' window. You can also drag and drop variables into the window you want.

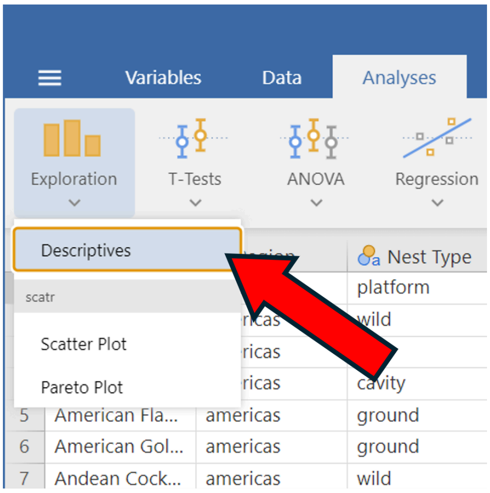


Figure 3.1: Step 2: Click on 'Exploration' and then 'Descriptives'

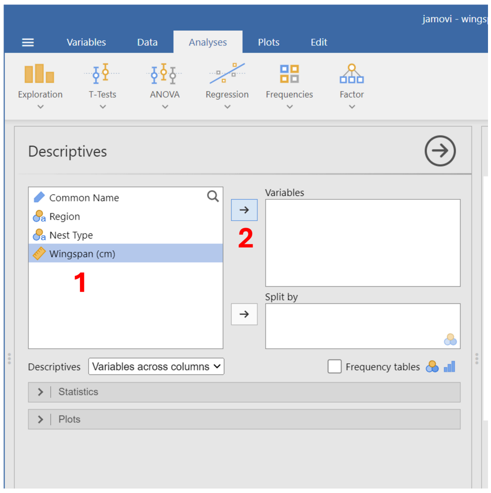
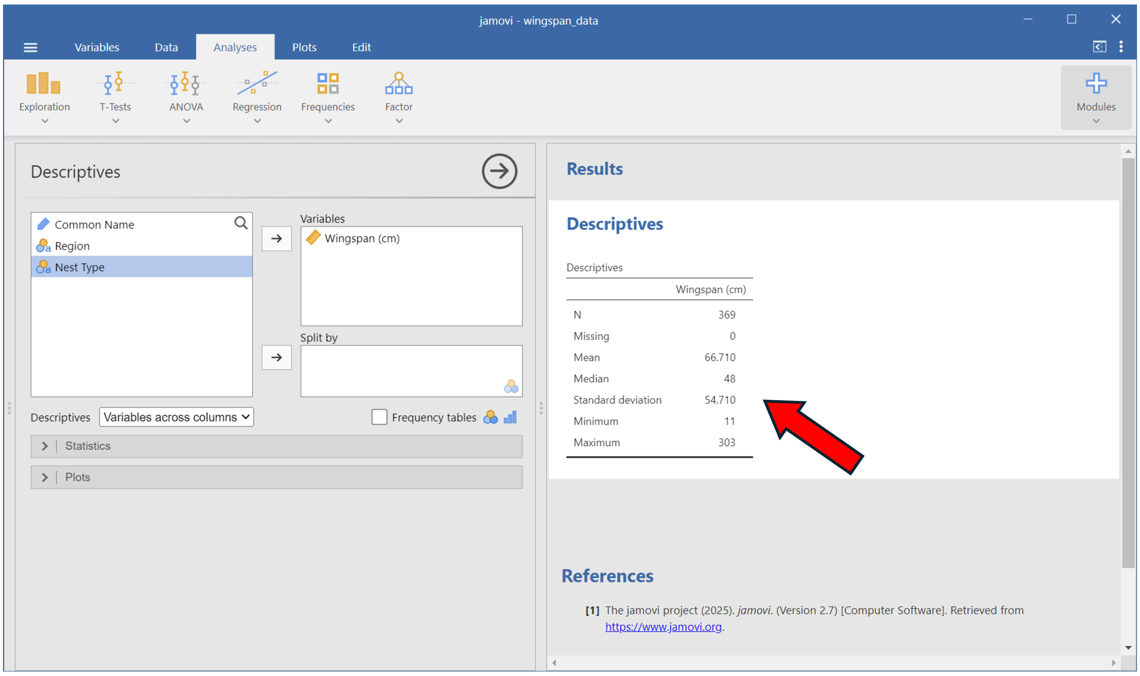


Figure 3.2: Step 3: Place the Wingspan variable into the 'Variables' window

After following these steps, the window on the right half of the screen should show these descriptive statistics:



3.2 Adding more statistics

By default, jamovi only shows the number of entries (N), the number of rows with missing values (Missing), the mean, median, standard deviation, minimum, and maximum. We can get other statistics by clicking on the 'Statistics' dropdown menu on the left half of the window:

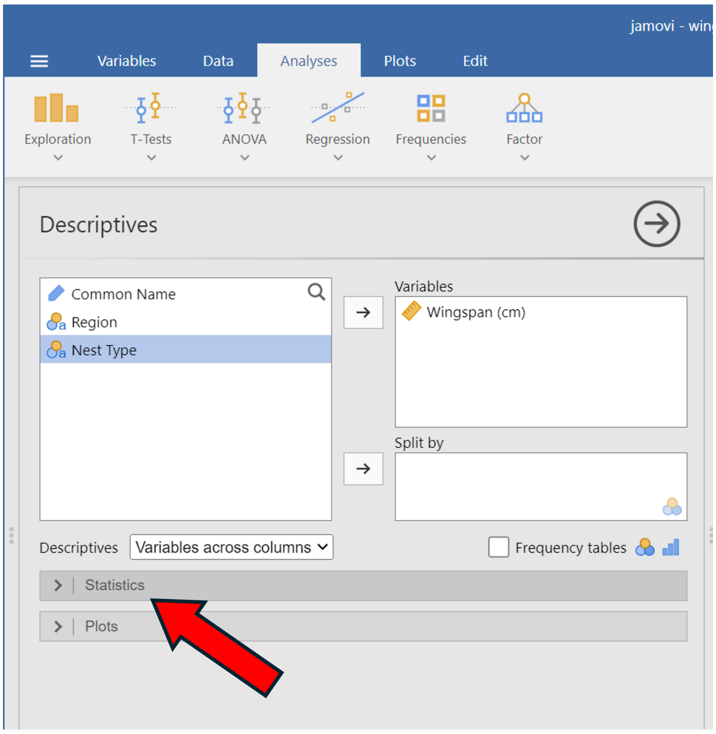


Figure 3.1: Click on 'Statistics' to add more statistics to the table

We can add or remove a number of statistics from this menu. Most of them we can ignore, but let's add the quartiles to our summary. This can be done by going to the **Percentile Values** headings and checking the 'Percentiles' box. By default, the values for the 'Percentiles' box should be '25, 50, 75'; if they aren't set to this, change them and then check the box. You should now have the 1st, 2nd, and 3rd quartiles in your descriptive statistics table:

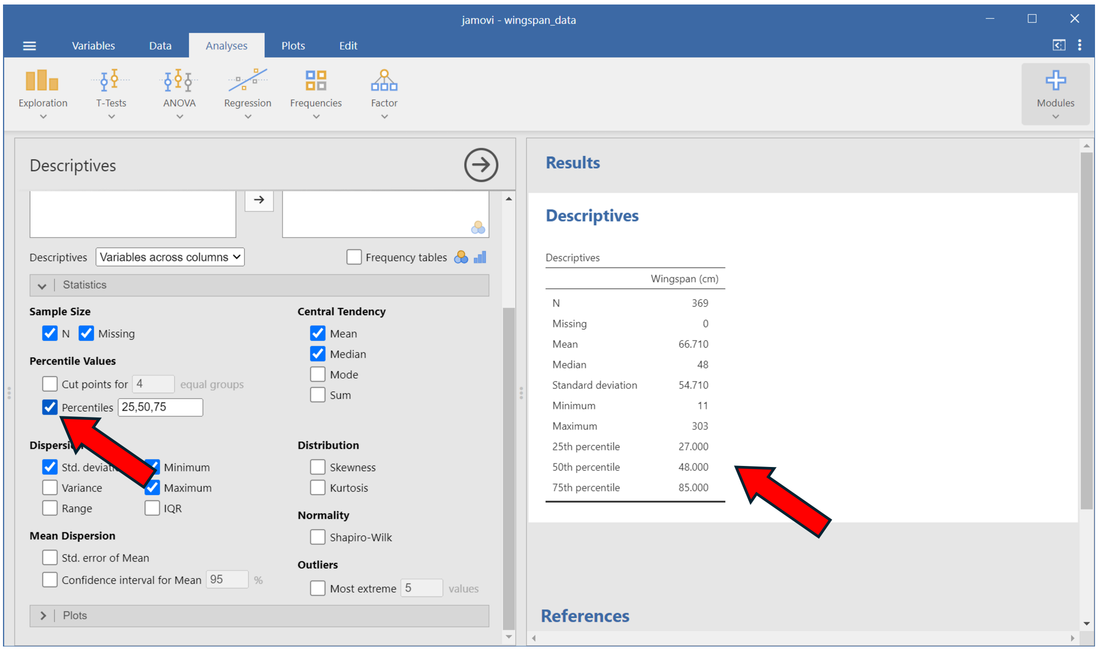


Figure 3.2: Adding quartiles to our table

3.3 Summary statistics split by group

Sometimes we want to look at summary statistics for particular groups. For example, we may want to compare the mean and median wingspan of birds from the European region to the mean and median wingspan of birds from Oceania region. To do this in jamovi, we can use the 'Split by' window.

Just like how we added the **Wingspan** variable to the 'Variables' window, add the **Region** variable to the 'Split by' window:

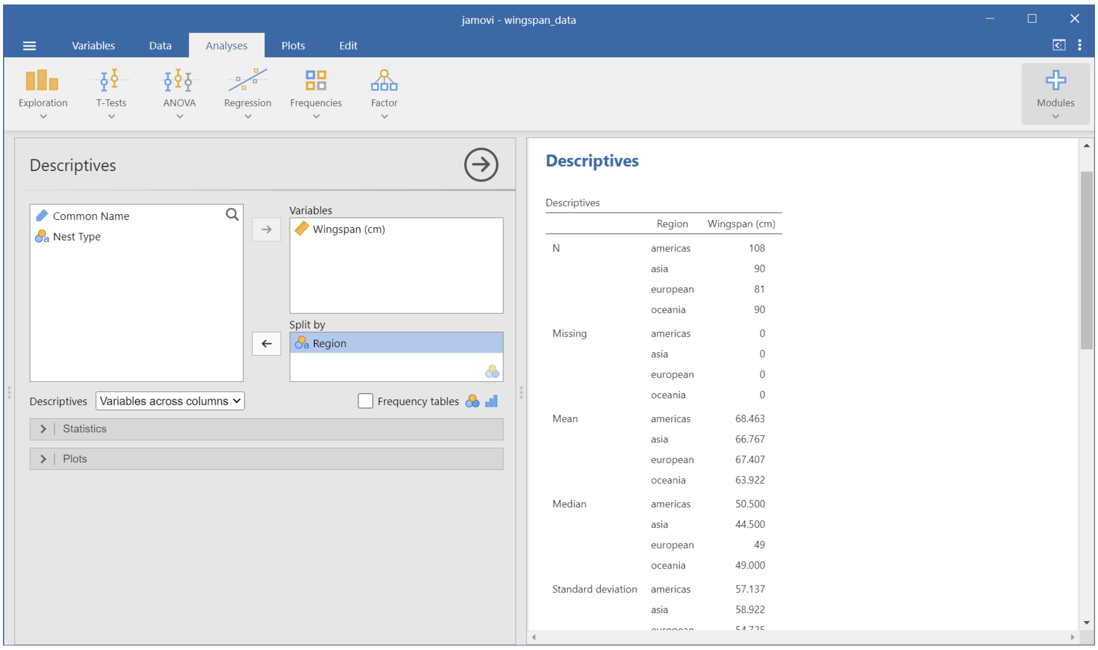


Figure 3.3: Adding **Region** to the 'Split by' window

On the right half of jamovi, we can see now that each of our statistics is now split up by region. However, it's a little difficult to read when the data is displayed vertically like this. Click on the 'Descriptives' drop down menu below the variable names and select 'Variables across rows' to group them horizontally instead of vertically.

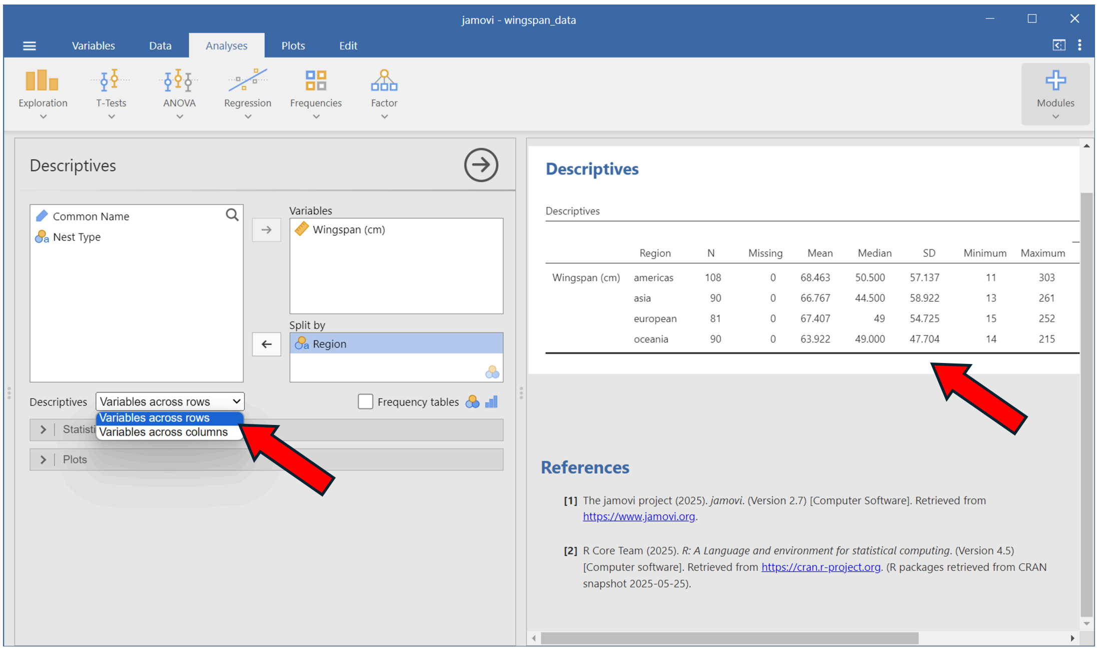


Figure 3.4: Clicking on 'Variables across rows' displays the statistics horizontally

In the next section, we'll show how to visualize distributions of data by creating tables and graphs.

4. Plots and Tables

In this section we'll go over how to generate 1-way and 2-way tables for categorical data, as well as how to generate histogram and boxplots for quantitative data.

4.1 Tables

One-Way Tables

First, we'll create a one-way table of regions.

1. Under 'Analyses', click on 'Frequencies' –> 'N Outcomes'
2. Add the **Region** variable to the 'Variables' window

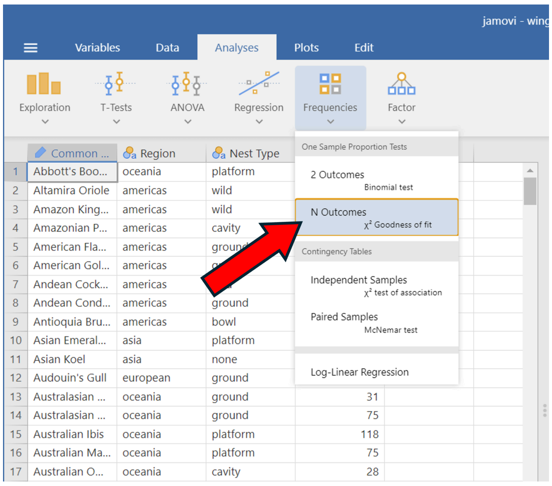


Figure 4.1: Click on 'N Outcomes'

Your jamovi settings and output should look like the following:

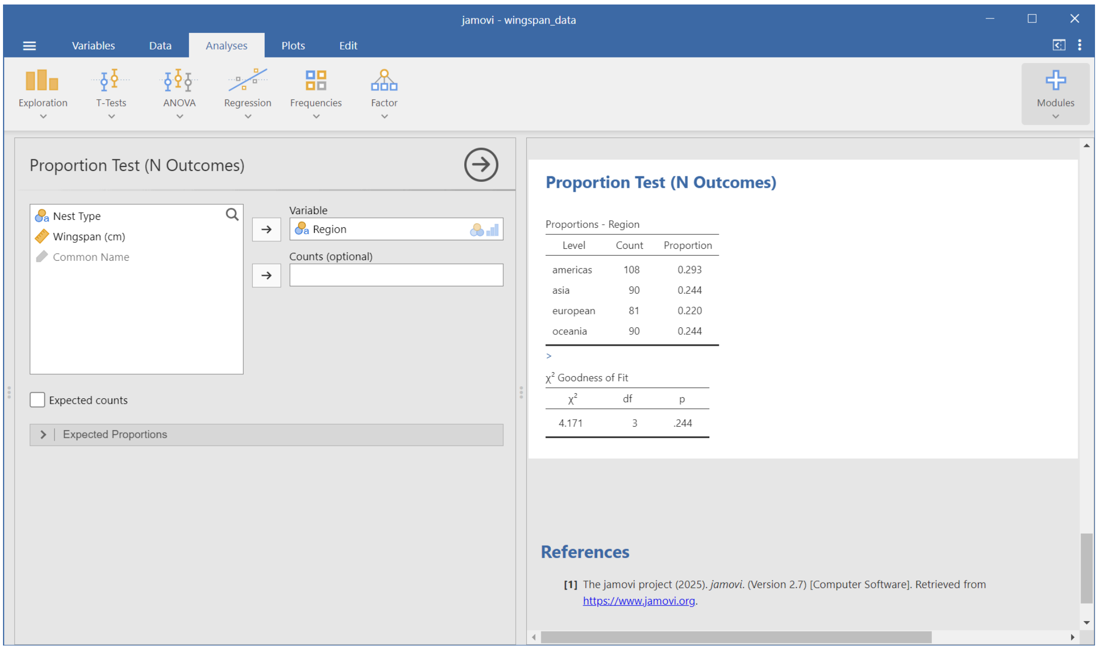


Figure 4.2: Output and settings for a one-way table

Two-Way Tables

Next, we'll create a two-way table (what jamovi calls a *contingency table*) of **Nest Type** vs **Region**.

1. Under 'Analyses', click on 'Frequencies' –> 'Independent Samples'
2. Add the **Region** variable to the 'Rows' window
3. Add **Nest Type** to the 'Columns' window

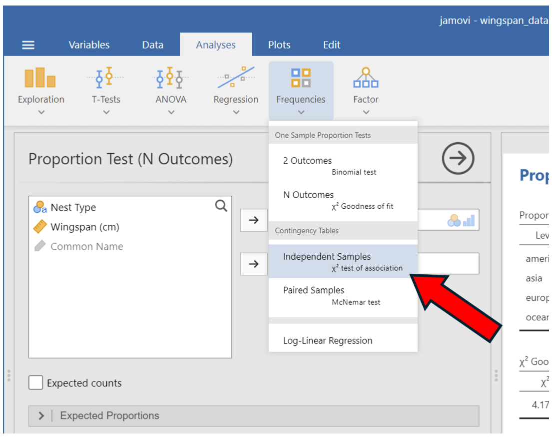


Figure 4.3: Click on 'Independent Samples'

Your jamovi settings and output should look like the following:

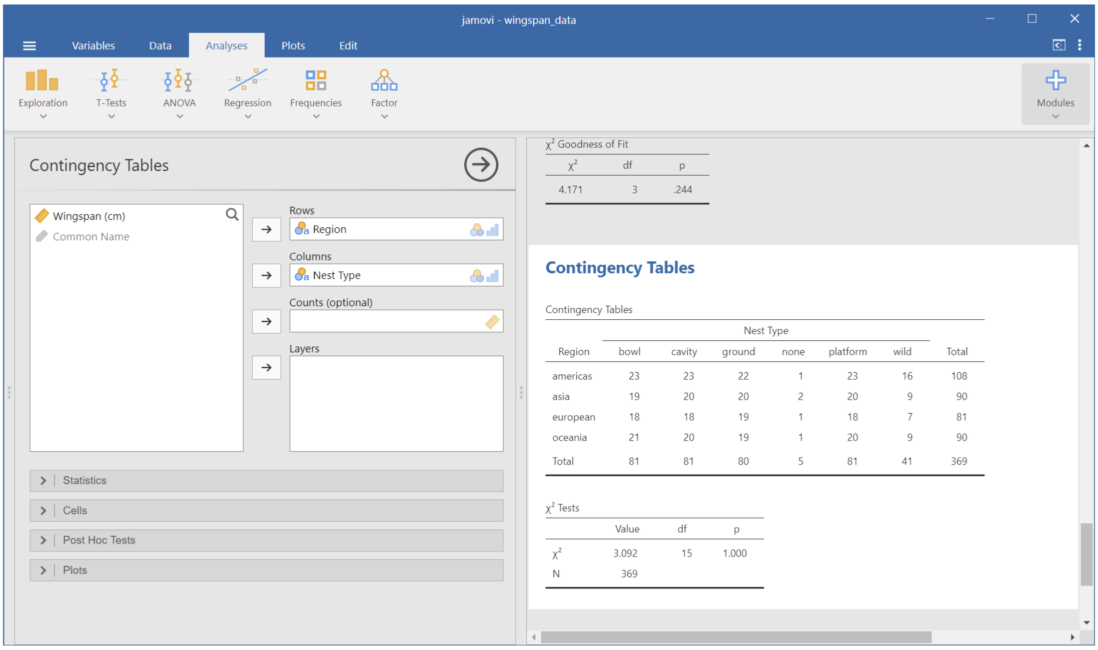


Figure 4.4: Output and settings for a two-way table

4.2 Plots

We're going to learn how to make histograms and boxplots of the **Wingspan** variable. To do so, we'll start with the following:

1. Under 'Analyses', click on 'Exploration' –> 'Descriptives'
2. Add the **Wingspan** variable to the 'Variables' window
3. Click on the 'Plots' dropdown menu below

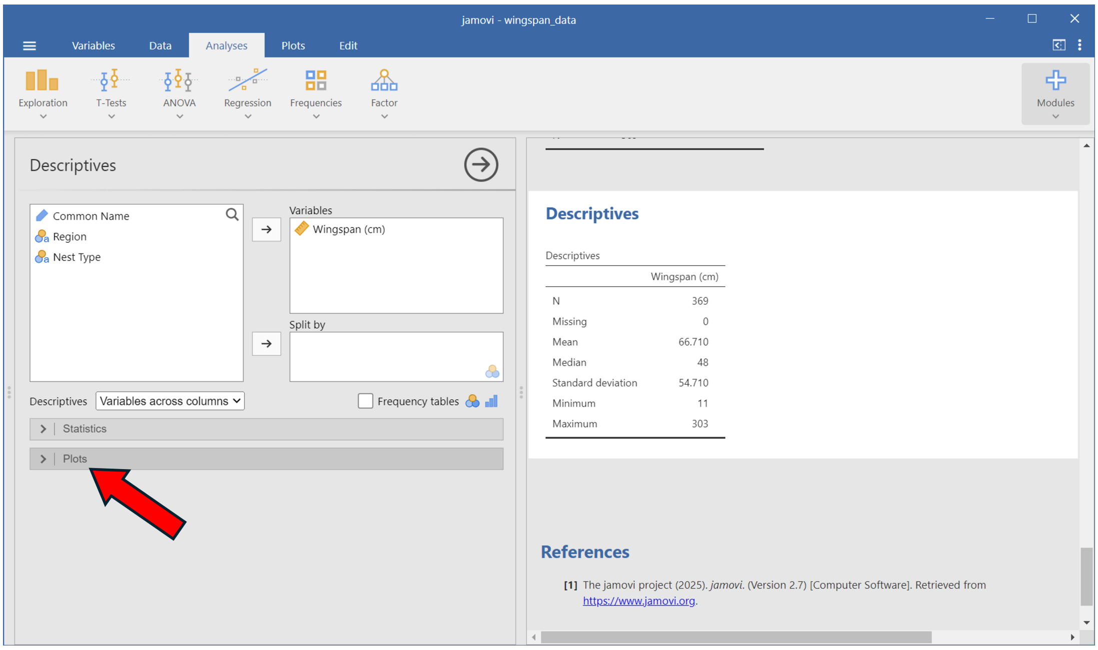


Figure 4.5: Click on the plots dropdown menu

Histograms

Now click on the 'Histogram' check box to create a histogram of wingspans.

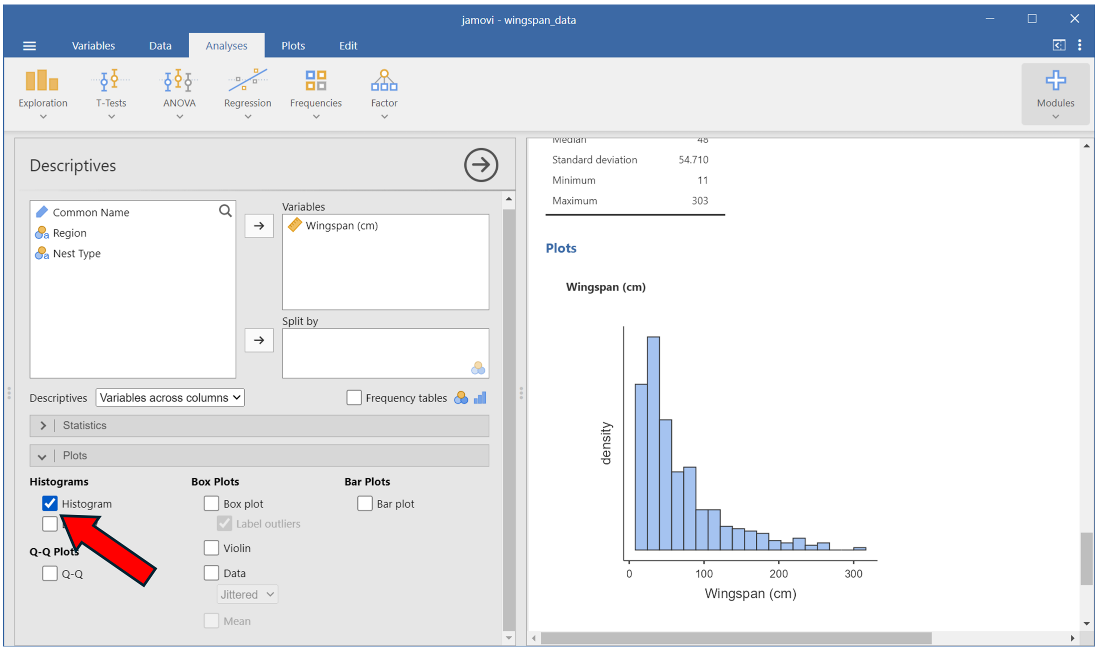


Figure 4.6: A single histogram of the **Wingspan** variable

It's also easy to split the histograms by another variable. We'll create a histogram of wingspans for each nest type by adding the **Nest Type** variable into the 'Split By' window:

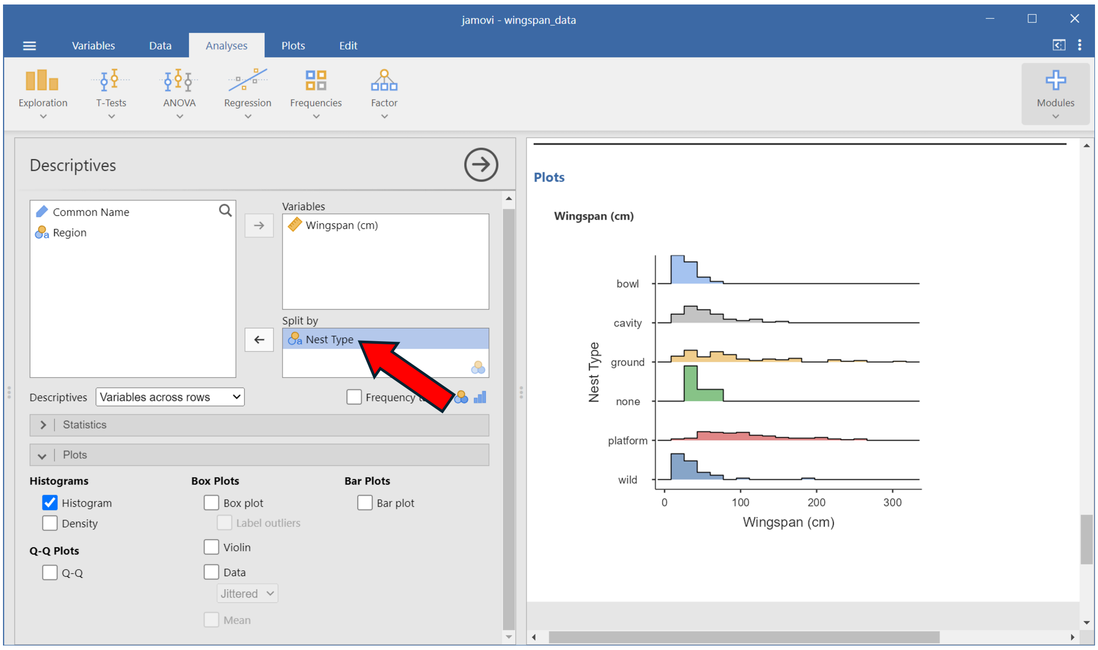


Figure 4.7: Histograms of the **Wingspan** variable, split by the **Nest Type** variable

Boxplots

Similarly, we can create boxplots by selecting the 'Boxplot' checkbox as seen below:

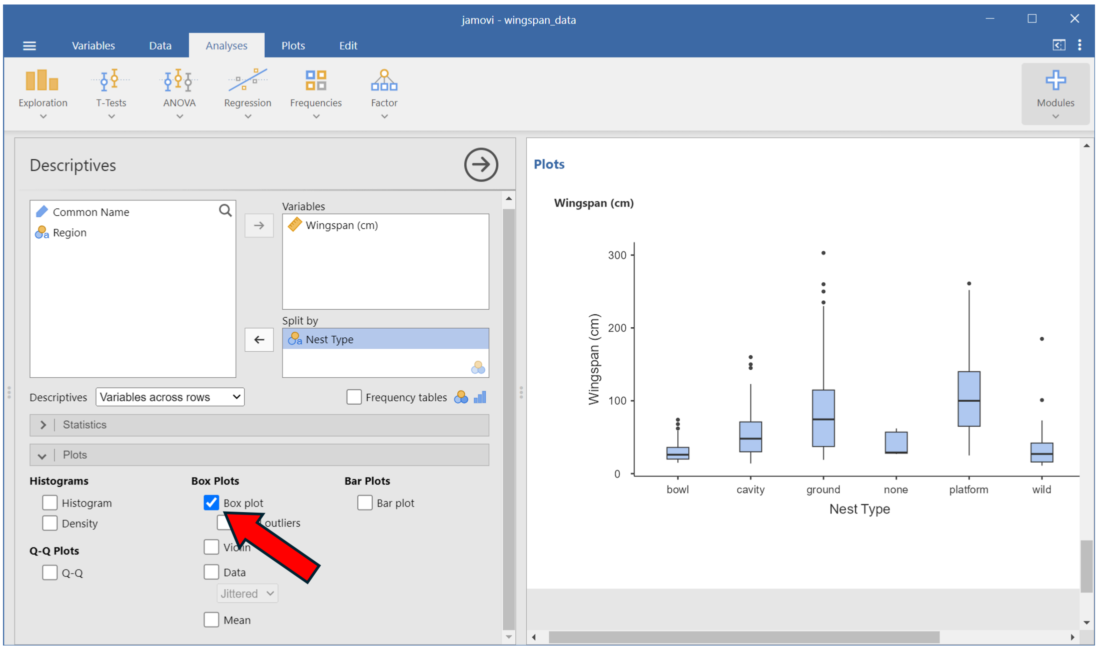


Figure 4.8: Creating boxplots

III Chapter 3: Doing Inference

5. T-Tests for One Population

In this section we'll learn how to compute t-statistics and confidence intervals for means in jamovi, including for paired differences. We'll use the **rotten_tomatoes.csv** dataset for this example, which has data on movies taken from RottenTomatoes.com. First, we'll calculate t-statistics and confidence intervals for the **Critic Rating** and **Audience Rating** variables individually. Then, we'll do a paired test on their difference.

Concept Check: Why would the difference between the Critics and Audience ratings be *paired*?

> Solution
>
> This is paired because each observation (each movie) is 'measured' *twice*: once by critics and once by the audience. We should expect to see a relationship between these two measurements. Note that this is a little trickier than our typical notion of a paired study, which usually involves a *before* and *after* measurement.

5.1 T-Tests for Movie Ratings

We're going to test the hypothesis that $H_0 : \mu = 60$ for both the **Critic Rating** and **Audience Rating** variables.

Start by loading in the **rotten_tomatoes.csv** dataset if you haven't already. Then:

1. Under Analyses, click 'T-Tests' –> 'One Sample T-Tests'
2. Add the **Critic Rating** and **Audience Rating** variables to the 'Dependent Variables' window.
3. Under **Tests**, ensure that 'Student's' is the only box checked (leave 'Bayes factor' unchecked)
4. Under **Hypothesis**, set 'Test value' to 60 and click '$\neq$ Test value'
5. Under **Additional Statistics**, check 'Mean difference' and 'Confidence interval' are checked, and set the confidence interval to 95%

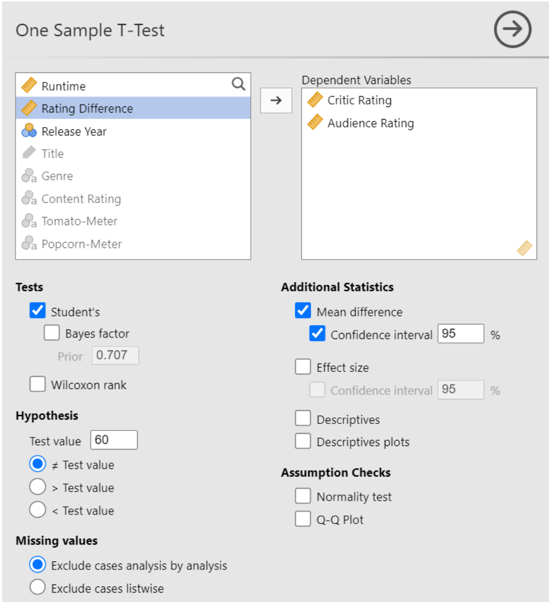
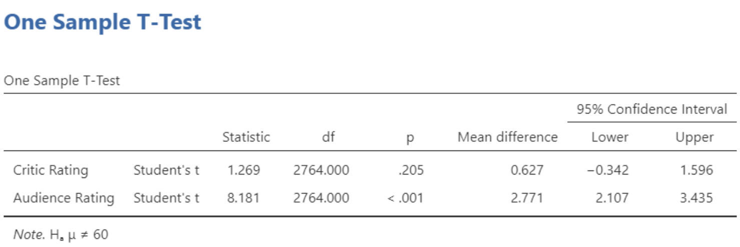


Figure 5.1: One sample t-test settings

You should get the following output:


One Sample T-Test

One Sample T-Test

		Statistic	df	p	Mean difference	95% Confidence Interval Lower	Upper
Critic Rating	Student's t	1.269	2764.000	.205	0.627	−0.342	1.596
Audience Rating	Student's t	8.181	2764.000	< .001	2.771	2.107	3.435

Note. H_a μ ≠ 60

Figure 5.2: One sample t-test output

5.2 Paired T-Test for Ratings Difference

Method 1 (No Ratings Differences column)

Paired t-tests look at the mean and standard error of the *differences* column. If we have the two columns but we do *not* have the differences column, we can use the following procedure:

1. Under 'Analyses', click on 'T-Tests' –> 'Paired Samples T-Test'
2. Add the **Critic Rating** variable and the **Audience Rating** variable to the 'Paired Variables' window. Note that the order will determine the sign of the statistic
3. Under **Tests**, ensure that 'Student's' is the only box checked (leave 'Bayes factor' unchecked)
4. Under **Hypothesis**, click 'Measure 1 ≠ Measure 2'
5. Under **Additional Statistics**, check 'Mean difference' and 'Confidence interval' are checked, and set the confidence interval to 95%

The final settings should look like so:

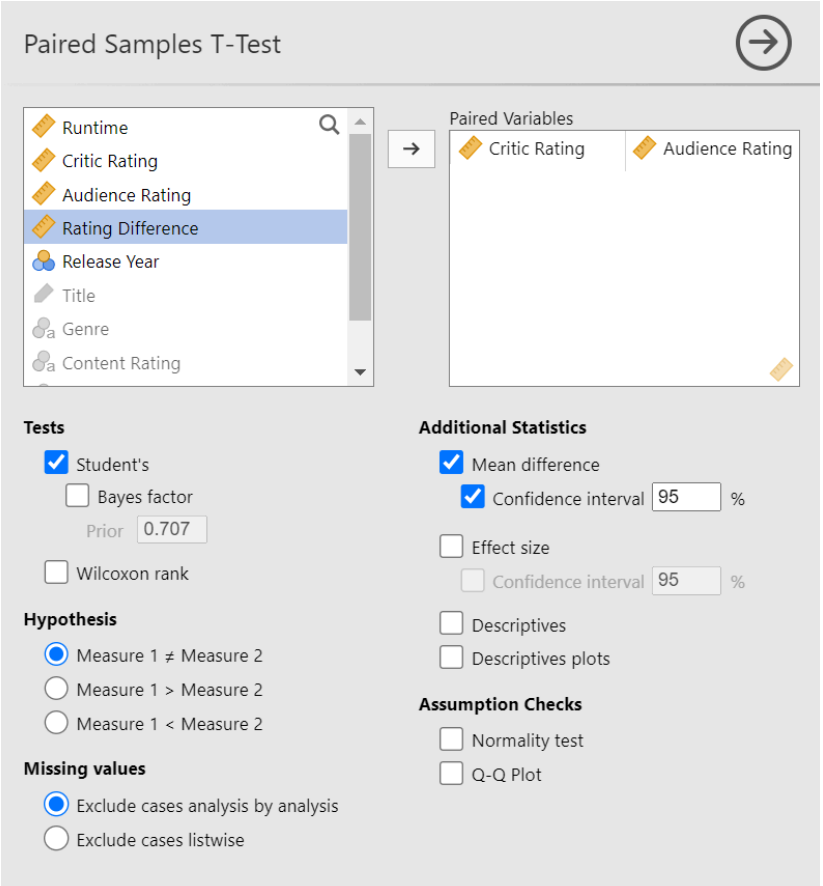


Figure 5.3: Paired t-test settings

And you should get the following table:

Paired Samples T-Test

Paired Samples T-Test

								95% Confidence Interval	
			statistic	df	p	Mean difference	SE difference	Lower	Upper
Critic Rating	Audience Rating	Student's t	−6.029	2764.000	< .001	−2.144	0.356	−2.841	−1.446

Note. H_a $\mu_{\text{Measure 1 - Measure 2}} \neq 0$

Figure 5.4: Paired t-test output

Method 2 (Differences column)

If a column is present that already has the differences between our two variables of interest, then we can just perform a one-sample t-test the same way we did in the previous section:

1. Under 'Analyses', click on 'T-Tests' –> 'One Sample T-Test'
2. Add the **Rating Difference** variable to the 'Dependent Variables' window
3. Under **Tests**, ensure that 'Student's' is the only box checked (leave 'Bayes factor' unchecked)
4. Under **Hypothesis**, set 'Test value' to 0 and click '≠ Test value'
5. Under **Additional Statistics**, check 'Mean difference' and 'Confidence interval' are checked, and set the confidence interval to 95%
6. Also under **Additional Statistics**, we need to check the 'Descriptives' box to get the standard error.

The final settings should look like so:

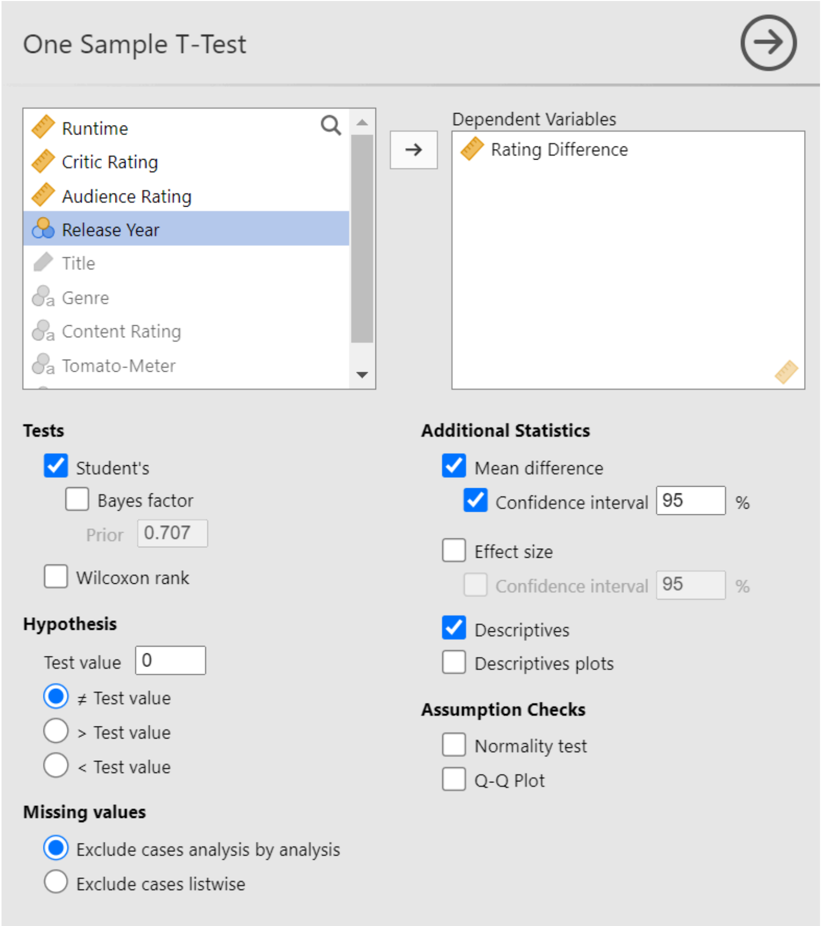
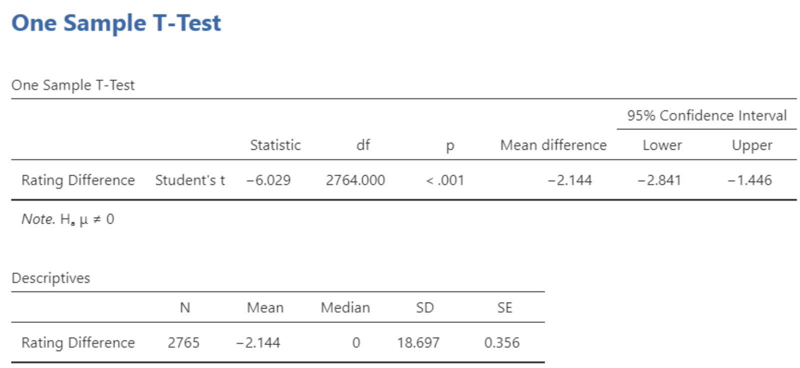


Figure 5.5: Paired t-test settings

And you should get the following table:



One Sample T-Test

One Sample T-Test

						95% Confidence Interval	
		Statistic	df	p	Mean difference	Lower	Upper
Rating Difference	Student's t	−6.029	2764.000	< .001	−2.144	−2.841	−1.446

Note. H_a $\mu \neq 0$

Descriptives

	N	Mean	Median	SD	SE
Rating Difference	2765	−2.144	0	18.697	0.356

Figure 5.6: Paired t-test output

6. T-Tests for Two Populations (Unpaired)

In this section we'll go over how to calculate t-statistics, standard errors, and confidence intervals for a difference of means problem with unpaired data. We're going to estimate the difference in mean arrival delay times between Delta flights and United flights from Denver to LA in 2024 using the **flight_data.csv** dataset.

In other words, we are going to test the hypothesis $H_0 : \mu_{Delta} = \mu_{United}$, where μ represents the population average arrival delay time from Denver to LA.

Start by loading in that data set. Then:

1. Go to the Analyses tab and click on 'T-Tests' –> 'Independent-Samples T-Test'
2. Add the **Delay** variable into the 'Dependent Variables' window
3. Add the **Airline** variable into the 'Grouping Variable' window
4. Ensure that the following boxes are checked:
    a. 'Student's'
    b. 'Mean Difference'
    c. 'Confidence Interval (95%)'
    d. 'Group 1 $\neq$ Group 2'

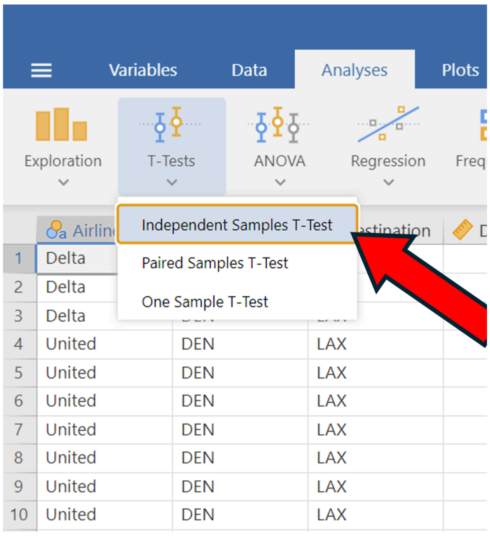


Figure 6.1: Select 'Independent-Samples T-Test'

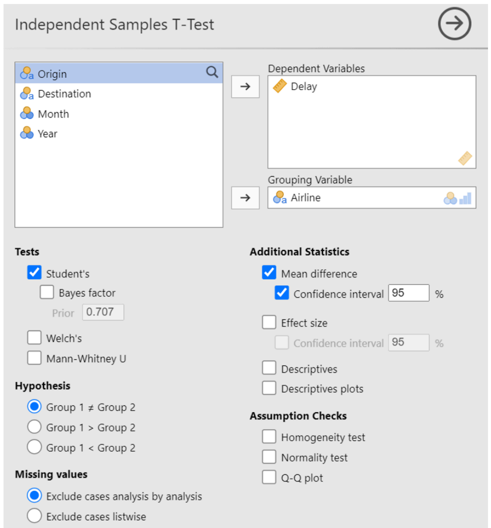


Figure 6.2: Final settings for the test

Your output should look like this:

Independent Samples T-Test

Independent Samples T-Test

		Statistic	df	p	Mean difference	SE difference	95% Confidence Interval Lower	95% Confidence Interval Upper
Delay	Student's t	−0.211 [a]	4229.000	.833	−0.393	1.857	−4.034	3.249

Note. H_a $\mu_{Delta} \neq \mu_{United}$

[a] Levene's test is significant (p < .05), suggesting a violation of the assumption of equal variances

Figure 6.1: Unpaired t-test output

7. Power Analysis

In this section we'll go over how to do power analysis in jamovi. There are two things to note in this section that differ from others we've done!

1. Power analysis isn't included in jamovi be default, so we'll need to download an additional **module**.
2. We don't need any data for this section!

Downloading the Power Module

1. Click on the 'Modules' –> 'jamovi library'.
2. Under the 'Available' tab, use the search bar and type in 'power'
3. Scroll until you see a module called 'jpower'.
4. Click the 'install' button.

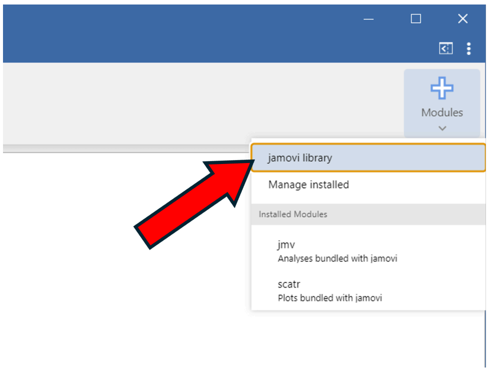


Figure 7.1: Click on 'Modules'

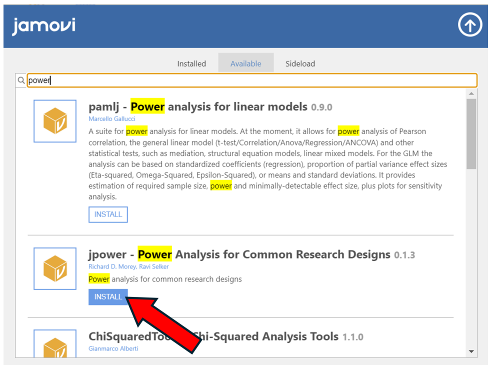


Figure 7.2: Search for the jpower module

If done correctly, you should now have a 'Power' section at the top of the screen in the 'Analyses' tab.

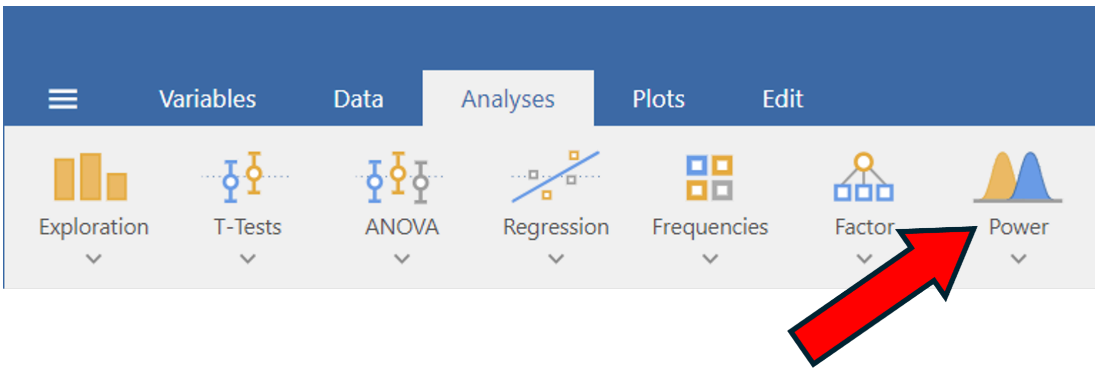


Figure 7.1: The menu after successfully installing the power module

Power Module settings.

Under the 'Analyses' tab, click on 'Power' –> 'Independent Samples T-test'. Make sure you have the following settings:

1. 'α (type I error rate)' = 0.05
2. 'Relative size of group 2 to group 1' = 1
3. Tails = 'two-tailed'

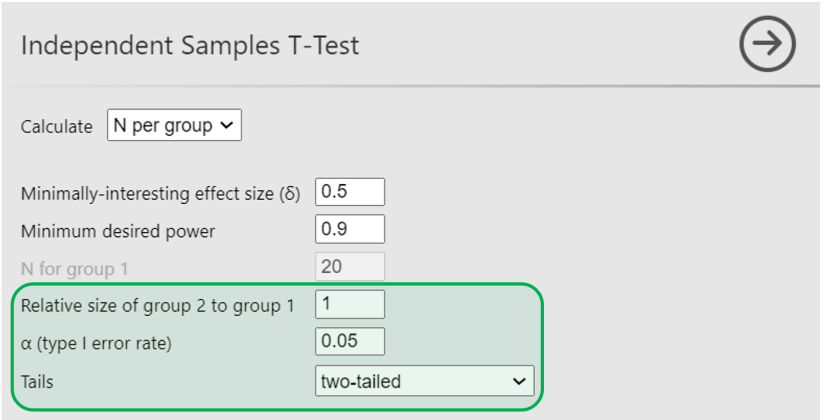


Figure 7.2: The required settings for power analysis

Power analysis problems can come in three different varieties:

1. Given a sample size and effect size, calculate power
2. Given power and an effect size, calculate the sample size
3. Given power and a sample size, calculate the effect size

Computing Power

For this problem, let's assume we have a sample size of 20 and an effect size of 0.5. To calculate power:

1. Click the dropdown menu next to 'Calculate' and select 'Power'
2. Set 'Minimally-interesting effect size (δ)' = 0.5
3. Set 'N for group 1' = 10

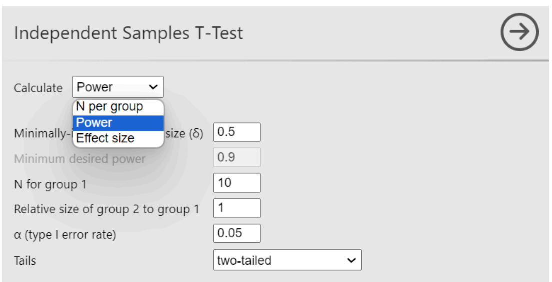


Figure 7.3: Settings for computing power

Your output should look like the following:

Independent Samples T-Test

The purpose of a *power analysis* is to evaluate the sensitivity of a design and test. You have chosen to calculate the sensitivity of the chosen design for detecting the specified effect size.

A Priori Power Analysis

	User Defined			
Power	N_1	N_2	Effect Size	α
0.185	10	10	0.500	0.050

Figure 7.4: Calculating power for $N = 20$, $\delta = 0.5$

The power given in the table is 0.185. That means that for a sample size of 20 and an effect size of 0.5, the probability of correctly rejecting a t-test is 0.185.

> ⚠ Note the sample size!
>
> If we want a total sample size of 20, we want 10 observations in each group. That's why we set $N = 10$ even though we stated at the beginning that the sample size of interest is 20.

Computing Sample Size

For this problem, we'll set our effect size to $\delta = 0.5$ and the power to 0.7, and we'll calculate the minimum sample size required.

1. Click the dropdown menu next to 'Calculate' and select 'N per group'
2. Set 'Minimally-interesting effect size (δ)' $= 0.5$
3. Set 'Minimum desired power' $= 0.7$

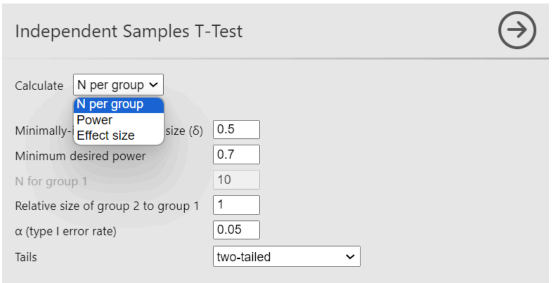
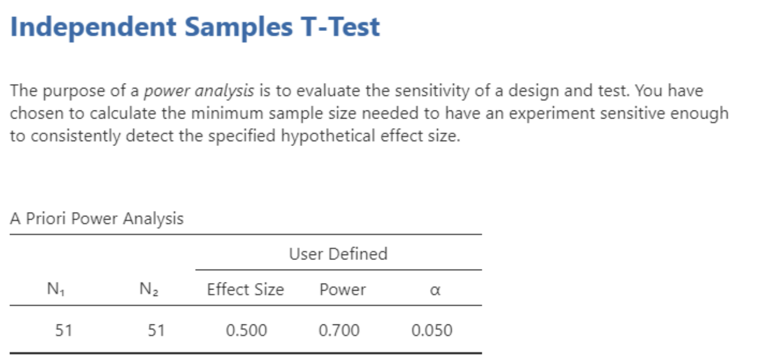


Figure 7.5: Settings for computing sample size

Your output should look like the following:



Independent Samples T-Test

The purpose of a *power analysis* is to evaluate the sensitivity of a design and test. You have chosen to calculate the minimum sample size needed to have an experiment sensitive enough to consistently detect the specified hypothetical effect size.

A Priori Power Analysis

		User Defined		
N_1	N_2	Effect Size	Power	α
51	51	0.500	0.700	0.050

Figure 7.6: Calculating sample size for power $= 0.7$, $\delta = 0.5$

The sample size required *for each group* is 51, so the *total* sample size required is $51 + 51 = 102$.

Computing Effect Size

Finally, we'll calculate effect size given sample size and desired power. We'll use a sample size of 20 and a power of 0.7

1. Click the dropdown menu next to 'Calculate' and select 'Effect size'
2. Set 'Minimum desired power' $= 0.7$
3. Set 'N for group 1' $= 10$

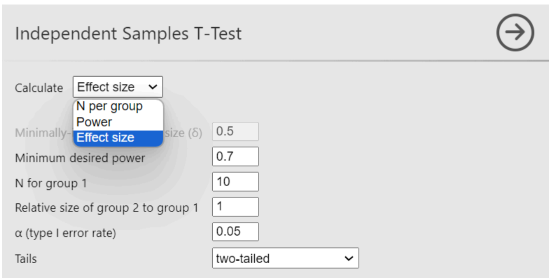
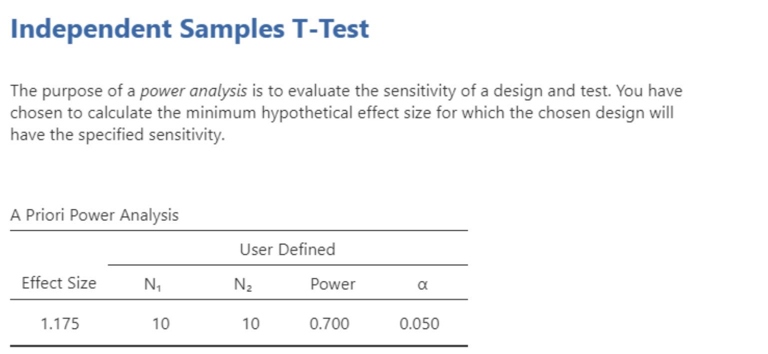


Figure 7.7: Settings for computing effect size

Your output should look like the following:



Independent Samples T-Test

The purpose of a *power analysis* is to evaluate the sensitivity of a design and test. You have chosen to calculate the minimum hypothetical effect size for which the chosen design will have the specified sensitivity.

A Priori Power Analysis

	User Defined			
Effect Size	N_1	N_2	Power	α
1.175	10	10	0.700	0.050

Figure 7.8: Calculating effect size for power $= 0.7$, $n = 20$

8. ANOVA and Multiple Comparisons

In this section we'll go over how to do ANOVA in jamovi. We're going to use the **rotten_tomatoes.csv** dataset to see if there is a difference in average runtime between different genres of movies.

1. In the 'Analyses' tab, click on 'ANOVA' –> 'ANOVA' (Note: do NOT click on 'One-Way ANOVA'. It won't have all of the information we need)
2. Add the **Runtime** variable to the 'Dependent variable' window
3. Add the **Genre** variable to the 'Fixed Factors' window

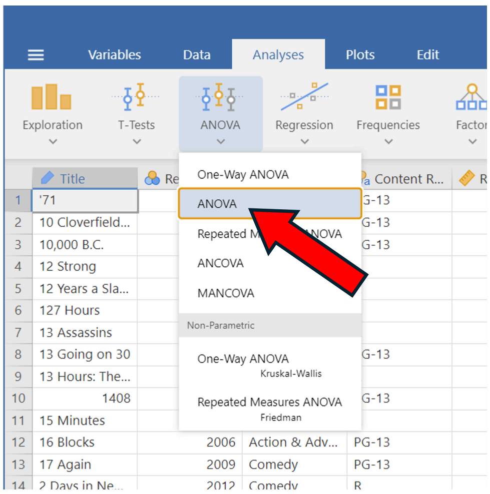


Figure 8.1: Click on 'ANOVA' –> 'ANOVA'

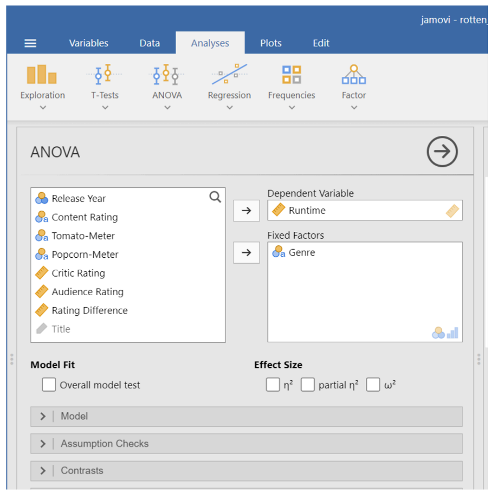


Figure 8.2: Add the variables

Your window should look something like the following. We use different notation than jamovi, so the image is labeled using 'our' notation:

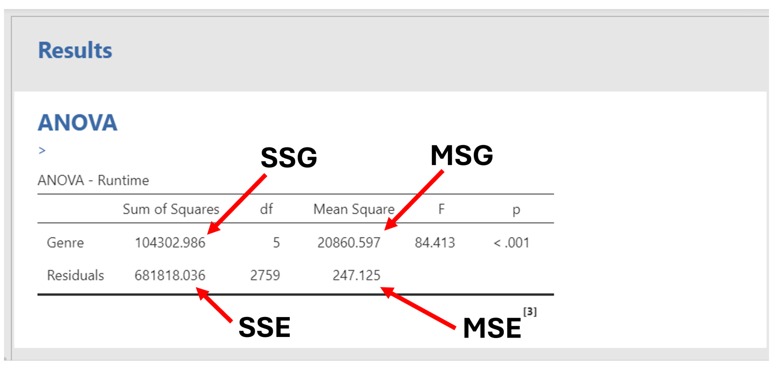


Figure 8.1: ANOVA table with out notation

When you see the word 'Residual', you can think of it as being synonymous with 'Error'.

8.1 Multiple Comparisons

After performing ANOVA and getting a p-value less than 0.05, we may want to then do multiple comparisons. To do this, stay on the ANOVA page from the previous section. Then:

1. Click on 'Post Hoc Tests' on the left side of the screen under the variable windows.
2. Place the **Genre** variable from the window on the left to the window on the right.
3. Under the **Correction** label, make sure the only box checked is 'No correction'.

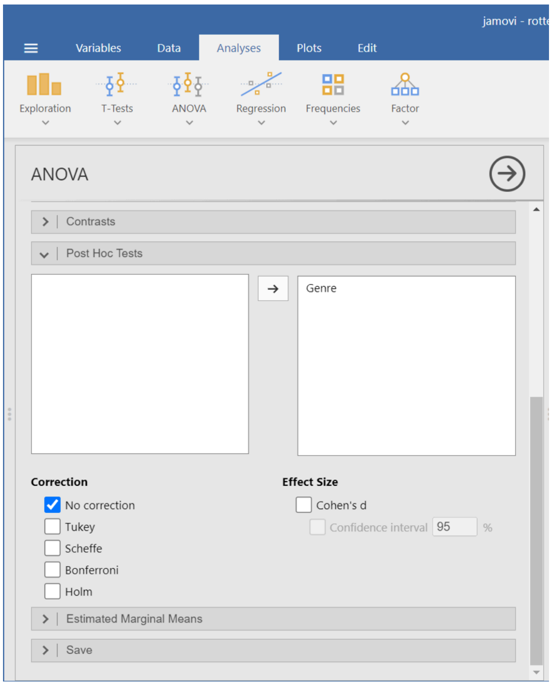


Figure 8.2: Multiple comparisons settings

After doing this, you should get the following table:

Post Hoc Tests

Post Hoc Comparisons - Genre

Comparison							
Genre		Genre	Mean Difference	SE	df	t	p
Action & Adventure	-	Animation	19.939	1.682	2759.000	11.855	< .001
	-	Comedy	10.328	0.785	2759.000	13.155	< .001
	-	Documentary	12.178	1.618	2759.000	7.528	< .001
	-	Drama	−0.471	0.773	2759.000	−0.609	.543
	-	Horror	14.281	1.618	2759.000	8.828	< .001
Animation	-	Comedy	−9.612	1.683	2759.000	−5.713	< .001
	-	Documentary	−7.762	2.198	2759.000	−3.531	< .001
	-	Drama	−20.410	1.677	2759.000	−12.170	< .001
	-	Horror	−5.659	2.198	2759.000	−2.574	.010
Comedy	-	Documentary	1.850	1.618	2759.000	1.143	.253
	-	Drama	−10.798	0.775	2759.000	−13.936	< .001
	-	Horror	3.953	1.618	2759.000	2.443	.015
Documentary	-	Drama	−12.649	1.613	2759.000	−7.843	< .001
	-	Horror	2.103	2.149	2759.000	0.978	.328
Drama	-	Horror	14.751	1.613	2759.000	9.147	< .001

Note. Comparisons are based on estimated marginal means

Figure 8.3: Multiple comparisons table

Interpreting the table

Focus on the 1st row of the table, highlighted in red in Figure 8.3.

This says that Action & Adventure movies are estimated to be 19.939 minutes longer than Animation movies, on average. The standard error for this estimate is 1.682 minutes, the t-statistic for this estimate is 11.855, and the p value is smaller than 0.001.

Now focus on the 7th row of the table, highlighted in green in Figure 8.3.

This says that Animation movies are estimated to be 7.762 minutes *shorter* than Documentary movies, on average. The standard error for this estimate is 2.198 minutes, the t-statistic for this estimate is −3.531, and the p value is smaller than 0.001.

9. Linear Regression

In this section we'll go over how to compute correlations and do simple and multiple linear regression in jamovi. We're going to predict the height of a tree using its diameter and species. Begin by loading in the **tree_data.csv** dataset from Canvas.

9.1 Correlation

Let's start by computing the correlation between the height and diameter of the trees.

1. Under 'Analyses' click on 'Regression' –> 'Correlation Matrix'
2. Add the **Diameter** and **Height** variables to the variables window.

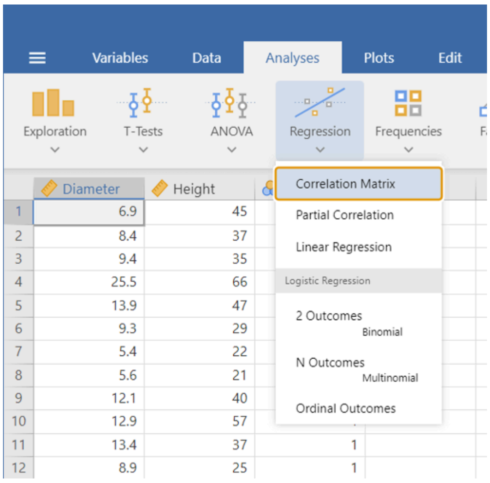


Figure 9.1: Select 'Correlation Matrix'

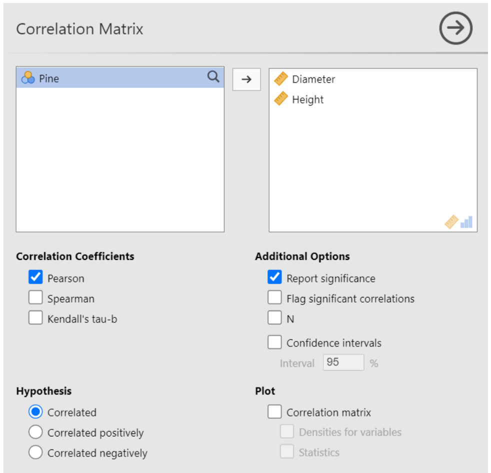


Figure 9.2: Add the **Diameter** and **Height** variables

You should get the following output:

Correlation Matrix

Correlation Matrix

		Diameter	Height
Diameter	Pearson's r	—	
	df	—	
	p-value	—	
Height	Pearson's r	0.774	—
	df	198	—
	p-value	< .001	—

Figure 9.1: Correlation matrix output

9.2 Simple Linear Regression

1. Under 'Analyses' tab, click on 'Regression' –> 'Linear Regression'.
2. Add the **Height** variable to the 'Dependent Variable' window.
3. Add the **Diameter** variable to the 'Covariates' window.

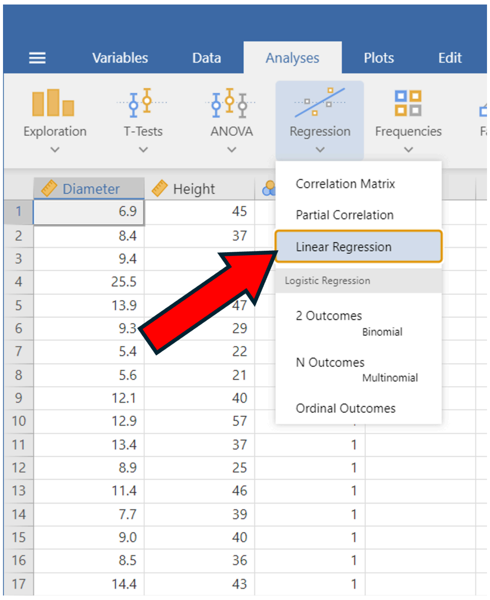


Figure 9.3: Select 'Linear Regression'

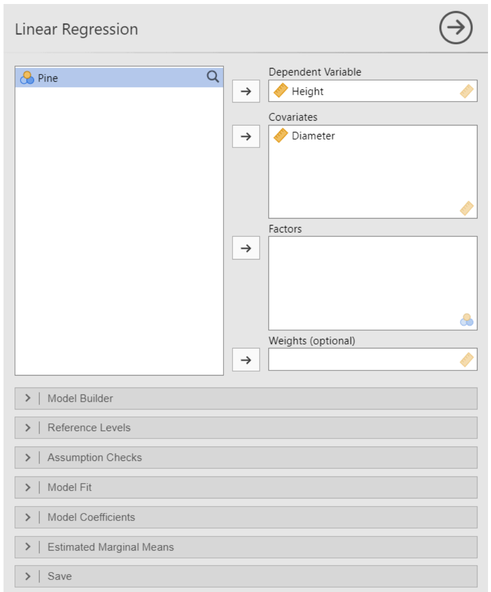


Figure 9.4: Add the **Height** and **Diameter** variables

That's it! Your output should match the image below:

Linear Regression

Model Fit Measures

Model	R	R^2
1	0.774	0.599

Note. Models estimated using sample size of N=200

Model Coefficients - Height

Predictor	Estimate	SE	t	p
Intercept	12.236	1.736	7.048	< .001
Diameter	2.659	0.155	17.185	< .001

Figure 9.2: Simple linear regression output

9.3 Multiple Linear Regression

Adding a Binary Variable

Now we're going to add the binary variable **Pine** which tells us if the observation is a Pine tree (1) or a Fir tree (0). Add the **Pine** variable to the 'Factors' window to get the following output:

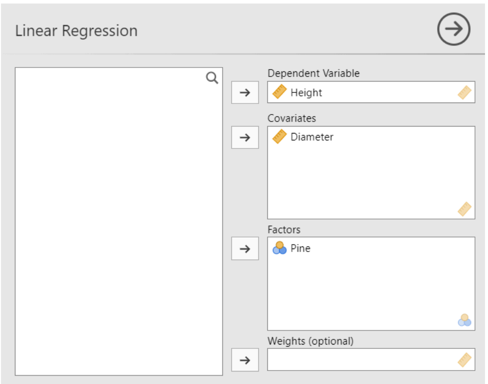


Figure 9.5: Adding a binary variable

Linear Regression

Model Fit Measures

Model	R	R^2
1	0.777	0.604

Note. Models estimated using sample size of N=200

Model Coefficients - Height

Predictor	Estimate	SE	t	p
Intercept ᵃ	12.883	1.775	7.257	< .001
Diameter	2.693	0.156	17.311	< .001
Pine:				
1 – 0	−2.018	1.254	−1.609	.109

ᵃ Represents reference level

Figure 9.6: Multiple linear regression output

Adding an Interaction Variable

Our **Pine** variable improved the fit, but was not statistically significant. It may be the case that Pine trees' diameters grow at a different rate relative to their height than Fir trees do. To test this, we're going to add an interaction between the **Diameter** variable and the **Pine** variable.

To do this, we need to create an interaction variable. Go back to the data and:

1. Double click on an empty data column to bring up a 'new variable' window
2. Click on 'New Computed Variable'
3. Name the variable 'Interaction' and in the window that has f_x next to it, type in 'Pine * Diameter'
4. Hit enter

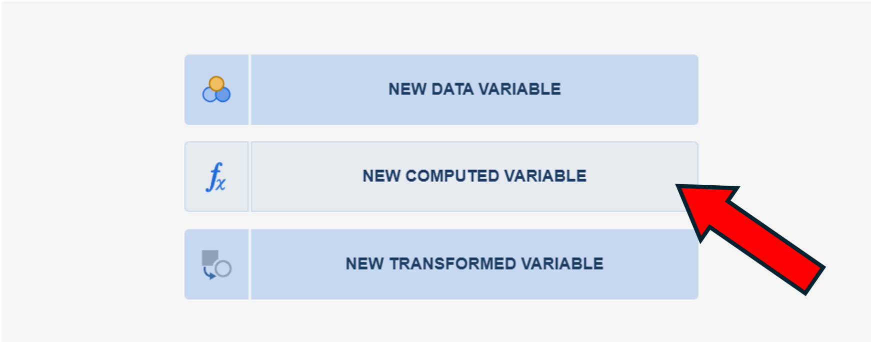


Figure 9.3: Double click on an empty column to open this window

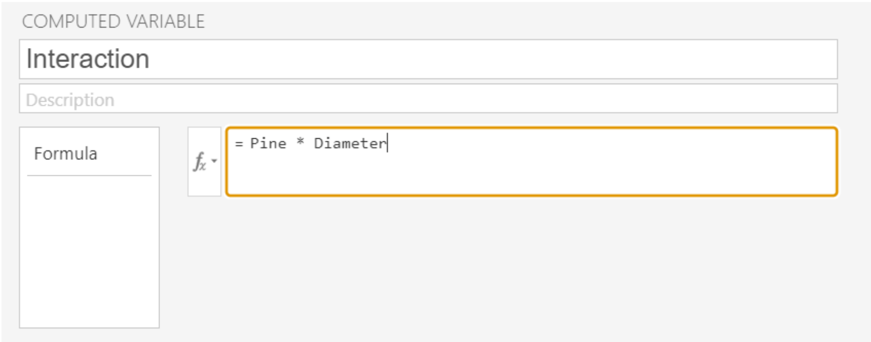


Figure 9.4: Create the new interaction variable

You should have a new column called **Interaction**. Now go back to the linear regression page and add the **Interaction** variable to the 'Covariates' window.

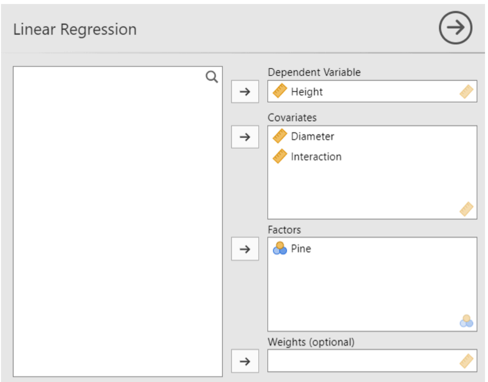
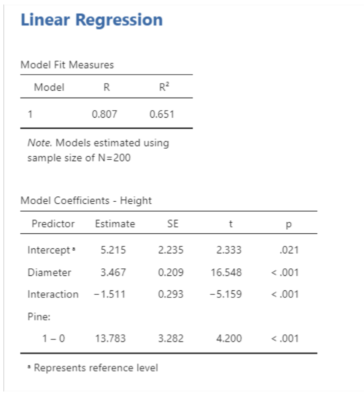


Figure 9.7: Adding the interaction variable


Linear Regression

Model Fit Measures

Model	R	R^2
1	0.807	0.651

Note. Models estimated using sample size of N=200

Model Coefficients - Height

Predictor	Estimate	SE	t	p
Intercept ᵃ	5.215	2.235	2.333	.021
Diameter	3.467	0.209	16.548	< .001
Interaction	−1.511	0.293	−5.159	< .001
Pine:				
1 – 0	13.783	3.282	4.200	< .001

ᵃ Represents reference level

Figure 9.8: Multiple linear regression with interaction output